Higher National Diploma in Information Technology

Second Year, First Semester Examination – 2018

HNDIT2312-Principles of Software Engineering

Instructions for Candidates:	No. of questions : 05
Answer Only **Four** questions	No. of pages : 02
All questions carry equal marks.	Time : **Two hours**

Q1.

i. Define the term "Software Engineering". (03 marks)

ii. What are the differences between Software engineering and the system engineering? (04 marks)

iii. What are the two types of Software products .Name them and explain. (04 marks)

iv. List out the six (06) Software Product Attributes. (06 marks)

v. Briefly explain four (04) characteristics of a software (08 marks)

(Total 25 Marks)

Q2.

i. List three (03) process models which provide a good visibility. (03 marks)

ii. Briefly explain why user requirements should be written using natural language. (04 marks)

iii. Explain briefly coupling and cohesion in modular design. (04 marks)

iv. Explain the use of coupling and cohesion in effective modular design. (06 marks)

v. List four (04) differences of functional and non-functional requirements of a system. (08 marks)

(Total 25 Marks)

Q3.

i. Write three(03) advantages of software architecture (03 marks)

ii. Give any four(04) architectural patterns (04 marks)

iii. List four (04) stages in testing process. (04 marks)

iv. Explain the difference between Process quality and Product quality. (06 marks)

v. Describe the difference between verification and validation in software checking process. (08 marks)

(Total 25 Marks)

Q4.

i. Briefly explain what is UML? (03 marks)
ii. Draw and name four(04) symbols in DFD Diagram (04 marks)
iii. When drawing DFD, you have to follow some rules and tips, Write four(04) of them (04 Marks)
iv. What is Use case Diagram? Name four (04) components in a Use case Diagram (06 marks)
v. Describe the four (04) best Practices for Designing an Interface (08 marks)

(Total 25 Marks)

Q5.

i. What is Project management? (03 marks)
ii. Describe the importance of change management in software maintenance. (04 marks)
iii. Explain the followings. (04 marks)
   a. Release management
   b. Configuration management planning
iv. Use below given table

Activity	**Dependencies**	**Duration**
A	**-**	**5**
B	**-**	**3**
C	**A, B**	**2**
D	**C**	**4**
E	**C**	**3**
F	**D**	**3**
G	**E**	**2**
H	**F, G**	**2**
I	**H**	**3**
J	**I**	**5**

   a. Draw a network diagram (08 Marks)
   b. Find critical path and minimum completion time of project (04 marks)
   c. If task "E" is delayed three (03) Days, what will happen to the critical path of the project? (02 marks)

(Total 25 Marks)